Airfoil Selection and Aerodynamic Analysis for Two RC Aircraft

Problem 1 technical report | Aircraft A: maximum payload with stable flight | Aircraft B: maneuverability (turns, rolls, loops)

1. Design Assumptions

Both aircraft are small electric RC models at **Re ≈ 2×10⁵** (chord ≈ 0.25 m, V ≈ 12 m/s, $\nu = 1.5\times10^{-5}$ m²/s). At this Re the boundary layer is mostly laminar, so laminar separation bubbles matter and behavior differs from full-scale data. Sea-level $\rho = 1.225$ kg/m³; free-stream turbulence is low; the wing is treated as 2D (no tip or fuselage effects) when comparing airfoils.

Table values are approximate literature figures (UIUC Low-Speed Airfoil Tests, Airfoil Tools/XFOIL polars) near Re ≈ 2×10⁵. They vary by source and transition model, so use them for *relative* comparison. The XFLR5 runs in Section 8 provide the like-for-like check.

2. Shortlisted Airfoils

Aircraft A (payload and stability)

1. **Clark Y**: flat-bottom classic for trainers and cargo models.
2. **NACA 4412**: moderate camber, widely documented.
3. **Selig S1223**: very high-lift, highly cambered low-Re profile.

Aircraft B (maneuverability)

1. **NACA 0009**: thin symmetric section.
2. **NACA 0012**: medium-thickness symmetric section.
3. **NACA 0015**: thick symmetric section.

3. Comparison Table (Re ≈ 2×10⁵, approximate)

Airfoil	Max thickness	Max camber	$C_{l,max}$	α_{stall}	$C_{d,min}$	$(L/D)_{max}$	$C_{m,0}$	Stall character
Clark Y	11.7 % @ 28 %c	3.5 %	≈ 1.2–1.3	≈ 12–14°	≈ 0.012	≈ 55–65	≈ −0.07	Gentle, progressive
NACA 4412	12 % @ 30 %c	4 % @ 40 %c	≈ 1.4–1.5	≈ 13–15°	≈ 0.011	≈ 55–65	≈ −0.10	Moderate, trailing-edge type
Selig S1223 (A)	**12.1 %**	**8.7 %**	**≈ 2.0–2.2**	**≈ 12–14°**	**≈ 0.017–0.02**	**≈ 65–75 (high C_l)**	**≈ −0.25 to −0.27**	**Fairly abrupt; needs care**
NACA 0009	9 % @ 30 %c	0 %	≈ 0.8	≈ 8–10°	≈ 0.009	≈ 40–45	0	Sharp (leading-edge type)
NACA 0012 (B)	**12 % @ 30 %c**	**0 %**	**≈ 0.85–0.95**	**≈ 9–11°**	**≈ 0.010**	**≈ 45–50**	**0**	**Moderately sharp**
NACA 0015	15 % @ 30 %c	0 %	≈ 0.95–1.0	≈ 11–13°	≈ 0.011	≈ 40–45	0	Gentler; more drag

Highlighted rows = selected profiles. L/D is for the section only and depends strongly on the C_l at which it is evaluated.

4. Selection and Justification

Aircraft A: Selig S1223

Payload is set by the maximum lift at a safe takeoff and stall speed: $L = \frac{1}{2}\rho V^2 S C_{L,max}$. For an illustrative wing of S = 0.5 m² with a 9 m/s stall speed, Clark Y ($C_{L,max} \approx 1.25$) carries about 31 N (≈ 3.1 kgf), whereas the S1223 ($C_{L,max} \approx 2.0$ at wing level, after 3D losses) carries about 50 N (≈ 5.1 kgf). That is roughly **60 % more gross weight** for the same wing area and stall speed, which is why high-lift sections like the S1223 dominate payload-class competitions.

The costs must be managed, not ignored:

- **Large nose-down moment** ($C_{m,0} \approx -0.26$): needs a larger horizontal tail volume (or longer tail arm) and a forward-biased CG to trim. Handled correctly, it also gives good static pitch stability.
- **Higher drag at low C_l**: cruise is less efficient than Clark Y, which is acceptable for a mission dominated by takeoff and climb.
- **Less forgiving stall**: about 2° of tip washout keeps aileron authority at stall.

> **Fallback: Clark Y.** If the rules favor docile handling over payload (novice pilot, tight CG envelope), Clark Y has a gentle stall and a small pitching moment.

Aircraft B: NACA 0012

A symmetric section suits aerobatics because the aircraft must fly upright, inverted and knife-edge with the same behavior:

- **Zero camber** gives $C_l = 0$ at $\alpha = 0$, so upright and inverted flight are equivalent (small trim change) and rolls are axial.
- **$C_{m,0} \approx 0$** and a nearly fixed aerodynamic center mean little trim change with speed or attitude, so control feel stays predictable through loops and snap maneuvers.
- **Low $C_{d,min}$ and 12 % thickness** retain energy through maneuvers while leaving depth for a spar to survive high-g pull-outs and for reliable aileron hinging.
- NACA 0009 has less drag but a sharper stall and little structural depth; NACA 0015 stalls more gently but pays a drag and weight penalty. 0012 is the balanced choice.

5. How Airfoil Geometry Affects Performance

Geometry parameter	Aerodynamic effect	Consequence for the aircraft
Camber	Shifts zero-lift angle negative, raises C_l at a given α and $C_{l,max}$; increases nose-down $C_{m,0}$	More payload and lower stall speed; more tail load and trim effort
Thickness	Thicker sections tolerate higher α and have a rounder leading edge (higher $C_{l,max}$, gentler stall) but add profile drag and, at low Re, can promote separation bubbles	Structural depth for spars and payload volume vs. drag
Position of max thickness/camber	Moves the suction peak and transition location	Affects drag-bucket width and stall type
Leading-edge radius	Large radius: better high-α behavior, gentle stall. Small radius: early leading-edge separation, abrupt stall	Stall predictability
Trailing-edge shape (flat bottom, reflex)	Reflex reduces nose-down moment; a flat bottom simplifies building and gives a good datum	Tail size, trim drag, build simplicity
Symmetry	Zero $C_{m,0}$ and symmetric $C_l(\alpha)$	Equal upright/inverted performance

In short: camber mostly sets *how much* lift, thickness mostly sets *how gently and robustly*, and leading-edge radius mostly sets *how the airfoil stalls*.

6. Can One Airfoil Serve Both Missions?

In principle one compromise section (e.g. Clark Y or NACA 4412) can fly both missions acceptably, but it is optimal for neither:

- **Aircraft A on a symmetric airfoil:** $C_{L,max}$ falls from about 2.0 to about 0.9, so for the same stall speed the wing area (or loading) must roughly double, adding weight and drag and cutting payload.
- **Aircraft B on S1223:** the large $C_{m,0}$ and positive-camber behavior make inverted flight very draggy and demand large elevator deflection; the abrupt stall and high drag at low C_l hurt snap rolls and energy retention.
- **Compromise (Clark Y / 4412):** $C_{l,max} \approx 1.2–1.5$ and manageable $C_{m,0}$, a good trainer or sport wing. Inverted flight needs noticeable down-elevator, and payload is below the S1223 case.
- **Engineering workaround:** flaps or flaperons that add camber for takeoff and retract for aerobatics partly bridge the gap, at a cost in complexity and weight.

The core trade-off is **lift capability and low stall speed** (camber, A) versus **symmetry, low drag and neutral pitching** (B).

7. Airfoil Characteristics vs. Whole-Aircraft Factors

The airfoil supplies only 2D section properties (C_l, C_d, C_m, stall type). Real behavior also depends on:

- **Wing loading (W/S) and aspect ratio:** set stall speed and turn radius ($V_{stall} = \sqrt{(2W / \rho S C_{L,max})}$); aspect ratio sets induced drag and the 3D lift-curve slope (below the 2D value).
- **Planform, twist, tip shape:** control spanwise lift distribution and where the wing stalls first.
- **Control surfaces and tail sizing:** the airfoil sets $C_{m,0}$, but tail volume, CG location, control area and throw set static stability, roll rate and pitch authority.
- **Structure and weight:** thickness limits spar depth; stiffness and strength set the maximum g, and so the minimum turn radius.
- **Thrust and power loading (T/W):** sets climb, vertical maneuvers and takeoff distance independent of the airfoil.
- **Fuselage, interference, surface finish:** add parasitic drag and disturb low-Re flow; a poor surface can negate small airfoil differences.

> An excellent airfoil on a badly balanced, heavy or underpowered airframe still performs poorly; a well-matched design often beats a "better" airfoil.

8. Bonus: XFLR5 Comparison of the Shortlisted Airfoils

Setup. All six sections analysed in XFLR5 (XFoil direct analysis), Mach 0, α from −4° to 18°, same settings for every foil. Assumed Re = 2×10^5 and $N_{crit} = 9$ (Section 1; adjust if your run differed). Curve colors:

Group	Colors → airfoil (assigned from curve behavior)
Aircraft B (symmetric)	white = NACA 0009 green = NACA 0012 blue = NACA 0015
Aircraft A (cambered)	yellow = Selig S1223 magenta = NACA 4412 red = Clark Y

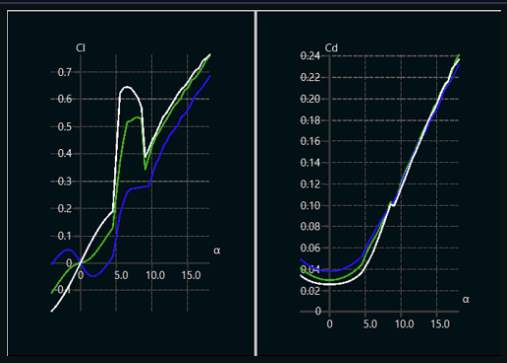


Fig. 1 – Aircraft B group. C_l–α (left) and C_d–α (right) for the three symmetric sections.

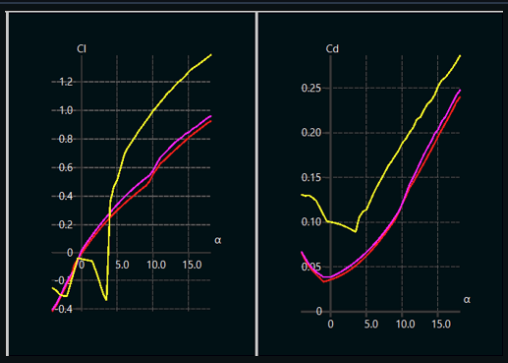


Fig. 2 – Aircraft A group. C_l–α (left) and C_d–α (right) for S1223, 4412 and Clark Y.

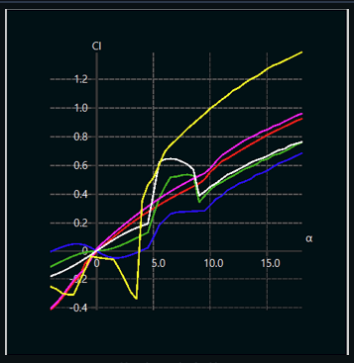


Fig. 3 – C_l vs α, all six airfoils.

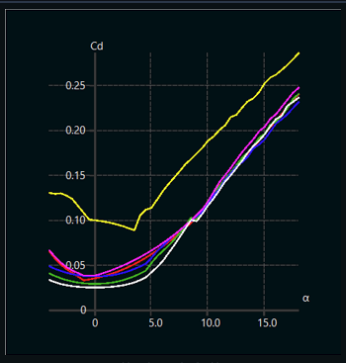


Fig. 4 – C_d vs α, all six airfoils.

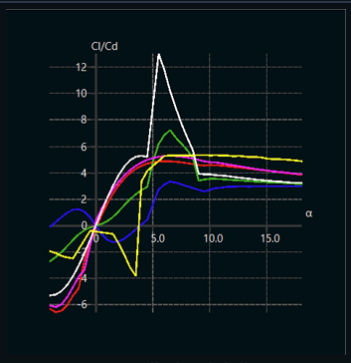


Fig. 5 – C_l/C_d vs α, all six airfoils.

Reading the plots (Figs. 1–5)

- **S1223 gives the most lift.** Its C_l is the highest of all six for α > 4° and reaches ≈ 1.4 at 18° with no stall inside the sweep, against ≈ 0.95 for 4412 and Clark Y and ≈ 0.65–0.77 for the symmetric group. This supports Aircraft A's choice.
- **S1223 pays in drag.** Its C_d is ≈ 0.09–0.10 near α = 0–4°, about 2–3× the others, and it stays highest across the sweep. Its C_l is also negative and erratic below ≈ 4°, which looks like a non-converged or separation-dominated region rather than real behavior, so treat α < 4° for this foil with caution.
- **Clark Y and 4412 are nearly identical** in C_l and C_d (4412 slightly higher), with C_l/C_d peaking near 5 at α ≈ 5–6° and falling to ≈ 4 at 18°.
- **Symmetric group (supports 0012 for B):** C_l passes through zero at α ≈ 0°, as expected. The thinnest foil (white) has the lowest C_d and the highest C_l/C_d peak (≈ 13 at α ≈ 5°), but it drops abruptly after ≈ 6° and shows a pronounced dip near 8–9°. The mid-thickness foil (green) peaks at ≈ 7 with a softer fall-off; the thickest (blue) has the lowest lift and peak C_l/C_d (≈ 3), consistent with the extra low-Re drag of thick sections.
- **Laminar separation bubbles.** The kinks and C_d steps near α ≈ 4–5° and 8–9° in the symmetric foils, and the plateau-then-jump in C_l, are the signature of bubble formation and bursting at this Re. Transition position drives these features, so they are sensitive to N_{crit} and Re.
- **Caveat.** The XFLR5 L/D values (≈ 3–13) are far below the literature section values in Section 3 (≈ 40–75). Section 3 comes from published polars at Re ≈ 2×10^5 taken near each foil's best C_l; the low L/D here points to a lower effective Re or conservative transition settings, plus the bubble effects above. Use Section 8 for *ranking* and Section 3 for magnitude. XFOIL also tends to over-predict $C_{l,max}$ after separation, so values near stall need care.

9. Conclusion

- **Aircraft A: Selig S1223** for maximum $C_{l,max}$ and payload, with a larger tail, forward CG and tip washout to preserve stability. Clark Y is the conservative fallback.
- **Aircraft B: NACA 0012**, a symmetric profile with neutral pitching, low drag and enough structural depth for aerobatics.
- One airfoil can fly both missions only as a compromise; final performance depends as much on wing loading, controls, structure and thrust as on the airfoil.

References

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